Homework #14

PHYS 4D: Modern Physics
Problems from Krane's textbook

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1 Questions, Chapter 11

1.1 Question 2

How should Eq. 11.7 be modified to be valid for MgO and BaO?

$$E_{\text{coh}} = \frac{1}{2}(B)(2N_A) = BN_A \tag{11.7}$$

1.1.1 Answer

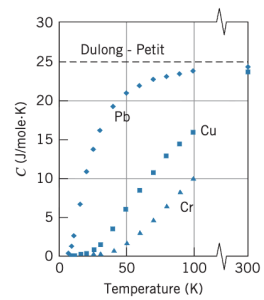
Let's take a closer look at that value B. It can be expanded.

$$B = -U(R_0) = \frac{\alpha e^2}{4\pi\varepsilon_0 R_0}\left(1 - \frac{1}{n}\right) \tag{1}$$

There are two values here to be explored. The first is the Madelung constant α, which is dependant on the binding structure. MgO and BaO are both fcc structures, so they have Madelung constant $\alpha = 1.7476$. The second is n, which comes from an exponent of the equilibrium separation determined from compressability data. For MgO, $n = 7$. For BaO, $n = 9.5$.

1.2 Question 3

From Figure 11.11 (on the right), estimate the Einstein temperature for lead. (Hint: Consider Eq. 11.11 when $T = T_E$.)



$$C = 3R\left(\frac{T_E}{T}\right)^2 \frac{e^{T_E/T}}{\left(e^{T_E/T}-1\right)^2} \tag{11.11}$$

1.2.1 Answer

Suppose we take the hint and set $T = T_E$.

$$C = 3R\left(\frac{T_E}{T_E}\right)^2 \frac{e^{T_E/T_E}}{\left(e^{T_E/T_E}-1\right)^2} = 3R\frac{e}{(e-1)^2} \tag{2}$$

The universal gas constant $R = 8.314\,\frac{\mathrm{J}}{\mathrm{mol{\cdot}K}}$, so $3R = 24.9\,\frac{\mathrm{J}}{\mathrm{mol{\cdot}K}}$.

$$C = \left(24.9\,\frac{\mathrm{J}}{\mathrm{mol}\cdot\mathrm{K}}\right)\frac{e}{(e-1)^2} = 22.9\,\frac{\mathrm{J}}{\mathrm{mol}\cdot\mathrm{K}} \tag{3}$$

Looking at the graph from Fig. 11.11, I'd estimate that that lands at around $\boxed{80\,\mathrm{K}}$.

1.3 Question 4

A graph of C/T vs. T^2 for solid argon (similar to Figure 11.13) goes through the origin; that is, its y-intercept is zero. Explain.

1.3.1 Answer

The value of C is bifurcated into the electron-based heat capacity as in (10.63) and the lattice-based heat capacity, as in the Debye equation for heat capacity in (11.12).

$$C_e = \frac{\pi^2 k^2 N T}{2E_F} \tag{10.63}$$

$$C_D = \frac{12\pi^4}{5} R\left(\frac{T}{T_D}\right)^3 \tag{11.12}$$

In this case for (10.63), N refers to the number of electrons contributed to the electron gas. Argon is a noble gas, so there are no electrons contributed to the gas and $N = 0$. This means that $C_e = 0$, so $C_e/T = 0$, which is why the intercept is zero.

1.4 Question 7

(a) Why does the electrical conductivity of a metal decrease as the temperature is increased? (b) How would you expect the conductivity of a semiconductor to change with temperature?

1.4.1 Answer (a)

We always assume that the lattice structure is roughly constant and ideal. This is not fully true. Higher temperatures increase the motion of the nuclei, and in turn increases the chance of a collision of an electron with a lattice ion. This results in higher resistivity at higher temperatures.

1.5 Answer (b)

I would expect the conductivity to lower as temperature increases for the reasons discussed in part (a).

1.6 Question 12

Three different materials have filled valence bands and empty conduction bands, and the Fermi energy lies in the middle of the gap. The gap energies are 10 eV, 1 eV, and 0.01 eV. Classify the electrical properties of these materials at room temperature and at 3 K.

1.6.1 Answer

10 eV is an insulator. 1 eV is a semiconductor. 0.01 eV is a conductor if not a superconductor.

2 Problem 7

Calculate the Coulomb energy and the repulsion energy for NaCl at its equilibrium separation.

2.1 Solution

According to Table 11.1, NaCl has an fcc structure. This means that the Madelung constant $\alpha = 1.7476$. The coulombic potential energy of an ion in an ionic solid is defined by equation (11.1).

$$U_C = -\alpha \frac{e^2}{4\pi\varepsilon_0} \frac{1}{R} \tag{11.1}$$

For NaCl, the nearest-neighbor separation $R_0 = 0.281\,\mathrm{nm}$. Plug all of these into equation (11.1).

$$U_C = -\alpha \frac{e^2}{4\pi\varepsilon_0} \frac{1}{R_0} = -1.7476 \frac{(8.99 \times 10^9\,\frac{\mathrm{N \cdot m^2}}{\mathrm{C^2}})(1.602 \times 10^{-19}\,\mathrm{C})^2}{0.281 \times 10^{-9}\,\mathrm{m}} \tag{4}$$

$$= \boxed{-1.435 \times 10^{-18}\,\mathrm{J}} \tag{5}$$

You can arrive at the repulsion energy by dividing the Coulomb energy by the negative of the exponent n. In the case of NaCl, $n = 9$ and $-n = -9$.

$$U_R = \frac{U_C}{-n} = \frac{-1.435 \times 10^{-18}\,\mathrm{J}}{-9} = \boxed{1.594 \times 10^{-19}\,\mathrm{J}} \tag{6}$$

3 Problem 11

(a) At what temperature do the lattice and electronic heat capacities of copper become equal to each other? Take $T_D = 343\,\mathrm{K}$ and $E_F = 7.03\,\mathrm{eV}$. (b) Which contribution is larger above this temperature? (c) Below this temperature?

3.1 Solution (a)

Copper has one valence electron, so for each mole of copper, $N = N_A$. From here, we can set (10.63) and (11.12) equal to each other and solve for T.

$$\frac{\pi^2 RkT}{2E_F} = \frac{12\pi^4}{5} R \left(\frac{T}{T_D}\right)^3 \tag{7}$$

$$\frac{kT}{E_F} = \frac{24\pi^2}{5} * \frac{T^3}{T_D^3} \tag{8}$$

$$T^2 = \frac{5kT_D^3}{24\pi^2 E_F} \tag{9}$$

$$T = \sqrt{\frac{5kT_D^3}{24\pi^2 E_F}} = \sqrt{\frac{5 * 8.617 \times 10^{-5}\,\mathrm{eV/K} * (343\,\mathrm{K})^3}{24\pi^2 * 7.03\,\mathrm{eV}}} \tag{10}$$

$$= \sqrt{\frac{17386\,\mathrm{eV}\cdot\mathrm{K}^2}{1665.2\,\mathrm{eV}}} = \sqrt{10.441\,\mathrm{K}^2} = \boxed{3.23\,\mathrm{K}} \tag{11}$$

3.2 Solution (b)

Calculate C in both cases with $T = 4\,\mathrm{K}$. I'll be dividing $\pi^2 R$ from both sides so as not to mix units and keep things simple.

$$\frac{C_e}{\pi^2 R} = \frac{kT}{2E_F} = \frac{8.617 \times 10^{-5}\,\mathrm{eV/K} * 4\,\mathrm{K}}{14.06\,\mathrm{eV}} = 2.45 \times 10^{-5} \tag{12}$$

$$\frac{C_D}{\pi^2 R} = \frac{12\pi^2}{5}\left(\frac{T}{T_D}\right)^3 = \frac{12\pi^2}{5}\left(\frac{4\,\mathrm{K}}{343\,\mathrm{K}}\right)^3 = 3.76 \times 10^{-5} \tag{13}$$

The lattice heat capacity is higher.

3.3 Solution (c)

Calculate C in both cases with $T = 3\,\mathrm{K}$. I'll be dividing $\pi^2 R$ from both sides so as not to mix units and keep things simple.

$$\frac{C_e}{\pi^2 R} = \frac{kT}{2E_F} = \frac{8.617 \times 10^{-5}\,\mathrm{eV/K} * 3\,\mathrm{K}}{14.06\,\mathrm{eV}} = 1.84 \times 10^{-5} \tag{14}$$

$$\frac{C_D}{\pi^2 R} = \frac{12\pi^2}{5}\left(\frac{T}{T_D}\right)^3 = \frac{12\pi^2}{5}\left(\frac{3\,\mathrm{K}}{343\,\mathrm{K}}\right)^3 = 1.58 \times 10^{-5} \tag{15}$$

The electronic heat capacity is higher.

4 Problem 12

(a) The heat capacity of solid argon at a temperature of 2.00 K is $2.00 \times 10^{-2}\,\frac{\mathrm{J}}{\mathrm{mol \cdot K}}$. What is the Debye temperature of solid argon? (See Question 4.)
(b) What value do you expect for the heat capacity at a temperature of 3.00 K?

4.1 Solution (a)

From Question 4, we know that the heat capacity of argon is equal to its Debye heat capacity.

$$C = \frac{12\pi^4}{5} R \left(\frac{T}{T_D}\right)^3 \tag{16}$$

$$T_D = T\sqrt[3]{\frac{12\pi^4 R}{5C}} = 2.00\,\mathrm{K}\sqrt[3]{\frac{12\pi^4 * 8.31}{5 * 2.00 \times 10^{-2}}} = 2.00\,\mathrm{K}\sqrt[3]{\frac{9713.6}{1.00 \times 10^{-1}}} \tag{17}$$

$$= 2.00\,\mathrm{K}\sqrt[3]{97136} = 2.00\,\mathrm{K} * 45.97 = \boxed{91.9\,\mathrm{K}} \tag{18}$$

4.2 Solution (b)

Use $T = 3.00\,\mathrm{K}$.

$$C = \frac{12\pi^4}{5} R \left(\frac{T}{T_D}\right)^3 = \frac{12\pi^4}{5} * 8.31 * \left(\frac{3.00}{91.9}\right)^3 = \boxed{6.76 \times 10^{-6}\,\frac{\mathrm{J}}{\mathrm{mol \cdot K}}} \tag{19}$$

5 Problem 13

When C/T is plotted against T^2 for potassium, the graph gives a straight line with a slope of $2.57 \times 10^{-3} \frac{\mathrm{J}}{\mathrm{mol \cdot K^4}}$. What is the Debye temperature for potassium?

5.1 Solution

First take the formula value of C/T, then divide that by T^2 to get the slope (C/T^3).

$$C = \frac{12\pi^4 R T^3}{5T_D^3} \tag{20}$$

$$\frac{C}{T} = \frac{12\pi^4 R T^2}{5T_D^3} \tag{21}$$

$$\frac{C}{T^3} = \frac{12\pi^4 R}{5T_D^3} = m \tag{22}$$

$$T_D = \sqrt[3]{\frac{12\pi^4 R}{5\left(\frac{C}{T^3}\right)}} = \sqrt[3]{\frac{12\pi^4 8.31}{5 * 2.57 \times 10^{-3}}\,\mathrm{K}^3} = \sqrt[3]{755925}\,\mathrm{K} = \boxed{91.1\,\mathrm{K}} \tag{23}$$

6 Problem 15

(a) In copper at room temperature, what is the electron energy at which the Fermi-Dirac distribution function has the value 0.1? (b) Over what energy range does the Fermi-Dirac distribution function for copper drop from 0.9 to 0.1?

6.1 Solution (a)

Here is the Fermi-Dirac Distribution Function.

$$f_{\mathrm{FD}}(E) = \frac{1}{e^{(E-E_F)/kT}+1} = 0.1 \tag{24}$$

$$e^{(E-E_F)/kT} + 1 = 10 \tag{25}$$

$$e^{(E-E_F)/kT} = 9 \tag{26}$$

$$\frac{E-E_F}{kT} = \ln(9) \tag{27}$$

$$E = kT\ln(9) + E_F \tag{28}$$

At room temperature, $kT \approx 0.025\,\mathrm{eV}$.An approximate equation for $E_F(T)$ is below.

$$E_F(T) \approx E_F(0)\left[1 - \frac{\pi^2(kT)^2}{12E_F(0)^2}\right] = E_F(0) - \frac{\pi^2(kT)^2}{12E_F(0)} \tag{29}$$

At $T = 0$, E_F is given by (30).

$$E_F(0) = \frac{h^2}{2m}\left(\frac{3N}{8\pi V}\right)^{2/3} \tag{30}$$

Looking at Example 11.4, the density of copper atoms at room tempera-

ture is $\frac{N}{V} = 8.49 \times 10^{28}$ atoms/m^3.

$$E_F(0) = \frac{(hc)^2}{2mc^2}\left(\frac{3}{8\pi}\frac{N}{V}\right)^{2/3} \tag{31}$$

$$= \frac{(1240\,\mathrm{eV}\cdot\mathrm{nm})^2}{2(0.511\times 10^6\,\mathrm{eV})}\left(\frac{3}{8\pi} * 8.49\times 10^{28}\mathrm{m}^{-3}\right)^{2/3} \tag{32}$$

$$= \frac{1.5376\times 10^{-12}\,\mathrm{eV}^2\mathrm{m}^2}{1.022\times 10^6\,\mathrm{eV}}\left(1.01\times 10^{28}\,\mathrm{m}^{-3}\right)^{2/3} \tag{33}$$

$$= 1.5045\times 10^{-18}\,\mathrm{eV}\cdot\mathrm{m}^2 * 4.683\times 10^{18}\,\mathrm{m}^2 = 7.0456\,\mathrm{eV} \tag{34}$$

$$E_F = 7.0456\,\mathrm{eV} - \frac{\pi^2(0.025\,\mathrm{eV})^2}{12 * 7.0456\,\mathrm{eV}} = 7.0455\,\mathrm{eV} \tag{35}$$

$$E = 0.025\,\mathrm{eV} * \ln(9) + 7.0455\,\mathrm{eV} = \boxed{7.1005\,\mathrm{eV}} \tag{36}$$

6.2 Solution (b)

Find the energy at $f_{\mathrm{FD}} = 0.9$.

$$f_{\mathrm{FD}}(E) = \frac{1}{e^{(E-E_F)/kT}+1} = 0.9 \tag{37}$$

$$e^{(E-E_F)/kT} + 1 = 90 \tag{38}$$

$$e^{(E-E_F)/kT} = 89 \tag{39}$$

$$\frac{E-E_F}{kT} = \ln(89) \tag{40}$$

$$E = kT\ln(89) + E_F = 0.025\,\mathrm{eV} * \ln(89) + 7.0455\,\mathrm{eV} = 7.1578\,\mathrm{eV} \tag{41}$$

The energy range is $\boxed{7.1578\,\mathrm{eV} \to 7.1005\,\mathrm{eV}}$.

7 Problem 16

Calculate the de Broglie wavelength of an electron with energy E_F in copper, and compare the value with the atomic separation in copper.

7.1 Solution

Recall the formula for the de Broglie wavelength.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \tag{42}$$

The Fermi energy fro copper os $7.03\,\mathrm{eV}$, while its mass is $511 \times 10^3\,\mathrm{eV}/c^2$.

$$\lambda = \frac{hc}{\sqrt{1.022 \times 10^6\,\mathrm{eV} * 7.03\,\mathrm{eV}}} = \frac{1240 \times 10^{-9}\,\mathrm{eV} \cdot \mathrm{m}}{\sqrt{7.184660 \times 10^6\,\mathrm{eV}^2}} \tag{43}$$

$$= \frac{1240 \times 10^{-9}\,\mathrm{eV} \cdot \mathrm{m}}{2680\,\mathrm{eV}} = \boxed{0.4626\,\mathrm{nm}} \tag{44}$$

The separation between atoms in copper is $0.289\,\mathrm{nm}$. This is about half the de Broglie wavelength of an electron in copper with energy E_F.

8 Problem 22

Estimate the ratio of the concentration of electrons in the conduction band of carbon (an insulator) and silicon (a semiconductor) at room temperature (293 K). The energy gaps are 5.5 eV for carbon and 1.1 eV for silicon. Assume the Fermi energy lies at the center of the gap.

8.1 Solution

From the Fermi-Dirac function, we have a distribution function.

$$f_{\mathrm{FD}}(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \tag{45}$$

As $e^{(E-E_F)/kT}$ increases (such as at room temperature), it approaches $f_{\mathrm{FD}} = e^{-(E-E_F)/kT}$. If we set E equal to the energy at the top of the gap, we can then say that $E - E_F = \frac{E_g}{2}$.

$$f_{\mathrm{FD}}(E) = e^{-\frac{E_g}{2kT}}; N = d\,f(E) \to N \propto e^{-\frac{E_g}{2kT}} \tag{46}$$

For the ratio of electron counts, just take a ratio. At room temperature, $kT \approx 0.025\,\mathrm{eV}$.

$$\frac{N_{\mathsf{C}}}{N_{\mathsf{Si}}} = \frac{e^{-\frac{E_{g;\mathsf{C}}}{2kT}}}{e^{-\frac{E_{g;\mathsf{Si}}}{2kT}}} = e^{-\frac{E_{g;\mathsf{C}}-E_{g;\mathsf{Si}}}{2kT}} = e^{-\frac{5.5\,\mathrm{eV}-1.1\,\mathrm{eV}}{2*0.025\,\mathrm{eV}}} \tag{47}$$

$$= e^{-\frac{4.4\,\mathrm{eV}}{0.050\,\mathrm{eV}}} = \boxed{6.05 \times 10^{-39}} \tag{48}$$

9 Problem 23

Estimate the ratio of the number of electrons in the conduction bands of germanium ($E_g = 0.66$ eV) and silicon ($E_g = 1.12$ eV) at a temperature of 400 K. Assume the Fermi energy is at the center of the gap.

9.1 Solution

For the ratio of electron counts, just take a ratio. See the previous problem.

$$kT = 8.617 \times 10^{-5}\,\mathrm{eV/K} * 400\,\mathrm{K} = 0.034468\,\mathrm{eV} \tag{49}$$

$$\frac{N_{\mathsf{Ge}}}{N_{\mathsf{Si}}} = \frac{e^{-\frac{E_{g;\mathsf{Ge}}}{2kT}}}{e^{-\frac{E_{g;\mathsf{Si}}}{2kT}}} = e^{-\frac{E_{g;\mathsf{Ge}}-E_{g;\mathsf{Si}}}{2kT}} = e^{-\frac{0.66\,\mathrm{eV}-1.12\,\mathrm{eV}}{2*0.034468\,\mathrm{eV}}} \tag{50}$$

$$= e^{\frac{0.46\,\mathrm{eV}}{0.068936\,\mathrm{eV}}} = \boxed{790.65} \tag{51}$$

10 Problem 28

The temperature of a sample of intrinsic silicon is increased by 100 K over room temperature (293 K). Estimate the increase in conductivity that we would expect from this increase in temperature. The gap energy in silicon is 1.1 eV.

10.1 Solution

Take the Fermi-Dirac function value for silicon at 293 K and 393 K.

$$f_{\mathrm{FD}}(293\,\mathrm{K}) = e^{-\frac{1.1\,\mathrm{eV}}{2*0.025\,\mathrm{eV}}} = 2.79 \times 10^{-10} \tag{52}$$

$$f_{\mathrm{FD}}(393\,\mathrm{K}) = e^{-\frac{1.1\,\mathrm{eV}}{2*8.617\times10^{-5}\,\mathrm{eV/K}*393\,\mathrm{K}}} = 8.84 \times 10^{-8} \tag{53}$$

There is a formula for the conductivity.

$$\sigma = \frac{ne^2\tau}{m} = \frac{ne^2}{m} * \frac{l}{v_{\mathrm{avg}}} \tag{54}$$

Do a ratio.

$$\sigma \propto T^{3/2} f_{\mathrm{FD}} \tag{55}$$

$$\frac{\sigma_f}{\sigma_i} = \left(\frac{T_f}{T_i}\right)^{3/2} \frac{f_{\mathrm{FD;f}}}{f_{\mathrm{FD;i}}} = \boxed{492} \tag{56}$$

11 Problem 29

(a) Estimate the fraction of the electrons in the valence band of intrinsic silicon that can be excited to the conduction band at a temperature of 100 K and at room temperature. Take the energy gap in silicon to be 1.1 eV. (b) By what factor would the conductivity of a metal change over this same temperature interval?

11.1 Solution (a)

Do another ratio.

$$n(100\,\mathrm{K}) = e^{-\frac{E_g}{2k*100\,\mathrm{K}}} = e^{-\frac{1.1\,\mathrm{eV}}{2*8.617\times10^{-5}*100\,\mathrm{K}}} = \boxed{1.9\times10^{-28}} \tag{57}$$

$$n(293\,\mathrm{K}) = e^{-\frac{E_g}{2k*293\,\mathrm{K}}} = e^{-\frac{1.1\,\mathrm{eV}}{2*8.617\times10^{-5}*293\,\mathrm{K}}} = \boxed{3.5\times10^{-10}} \tag{58}$$

11.2 Solution (b)

$$\frac{\sigma_f}{\sigma_i} = \left(\frac{T_f}{T_i}\right)^{3/2}\frac{n_f}{n_i} = \boxed{4.23\times10^{18}} \tag{59}$$